

Follow one record

Main Task 1 | Complete this sheet or the matching digital table.

Name: _____ Class: _____ Date: _____

Open activity.txt as text and inspect lines 1, 2 and 14. In fitness_working.py, find def LoadActivity. Read and trace by hand; no program run is needed. Use this sheet for the table and all three questions, then add a photo to the lesson notebook.

The current person has ID "1001". Python indexes start at zero: [0] is the ID, [3] the week and [7] the distance. Compare each record's ID with 1001, then decide if append executes. Records 3-13 are omitted below; the program still reads them.

```

124 def LoadActivity(Individual, ActivityLog):
125     ActivityLog = []
126     DataFile = open(ACTIVITY_FILE, 'r')
127     DataLine = DataFile.readline()
128     while DataLine != "":
129         Activity = Exercise()
130         ActivityData = DataLine.split(",")
131         Activity.PersonID = ActivityData[0]
132         Activity.Age = int(ActivityData[1])
133         Activity.DayNo = int(ActivityData[2])
134         Activity.WeekNo = int(ActivityData[3])
135         Activity.Time = ActivityData[4]
136         Activity.Type = ActivityData[5]
137         Activity.Duration = int(ActivityData[6])
138         Activity.Distance = float(ActivityData[7])
139         Activity.AverageHeartRate = int(ActivityData[8])
140         Activity.Weight = float(ActivityData[9])
141         Activity.CaloriesBurned = CalculateCalories(Activity, Individual)
142         if Activity.PersonID == Individual.ID:
143             ActivityLog.append(Activity)
144         DataLine = DataFile.readline()
145     DataFile.close()
146     return ActivityLog

```

Selected records from activity.txt

1: 1001,21,1,3,08:30,Running,60,10.0,148,70.1
 2: 1001,21,2,3,07:30,Running,58,10.9,137,71.2
 14: 1002,60,6,5,10:20,running,40,20.0,120,90.0

Record	Person ID read	Week	Distance	Matches ID 1001? True / False	Added to list? Yes / No
1 (first)					
2 (second)					
14 (final)					

1. Which condition decides whether a record enters ActivityLog? What would happen if that check were removed?

2. What does LoadActivity return to its caller?

3. Find AddActivity and AppendActivityToFile in your IDE. Name the statements that update the list and save the file. Explain how the two actions differ.
